```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error
from sklearn.ensemble import RandomForestRegressor
```

```python
path = r"D:/Projet 4/TaxiOut AI/data/featured_data.parquet"

df = pd.read_parquet(path)

print(df.shape)
```

```
(339013, 28)
```

------------------------TARGET taxi_out---------------------------

```python
y = df["taxi_out"]
```

-------------------------FEATURES----------------------

```python
X = df.drop(columns=["taxi_out"])
```

----------------VERIFIER LES TYPES------------------------

```python
print(X.dtypes)
```

```
month                                int32
day_of_month                         int32
scheduled_departure_time             int32
scheduled_arrival_time               int32
unique_carrier              string[python]
flight_number                        int32
scheduled_elapsed_time             float64
departure_delay                    float64
origin                      string[python]
dest                        string[python]
distance                             int64
cancelled                            int32
diverted                             int32
carrier_delay                      float64
weather_delay                      float64
nas_delay                          float64
security_delay                     float64
late_aircraft_delay                float64
year                                 int32
day                                  int32
day_of_week_num                      int32
is_weekend                           int64
departure_hour                       int32
is_peak_hour                         int64
airport_traffic                      int64
carrier_traffic                      int64
short_flight                         int64
dtype: object
```

--------------------------------------Encoding---------------------------------

```python
X = pd.get_dummies(X, drop_first=True)
print(X)
```

```
        month  day_of_month  scheduled_departure_time  scheduled_arrival_time  \
0           1             4                      1125                    1240
1           1             4                       545                     880
2           1             4                       875                    1030
3           1             4                       415                     570
4           1             4                      1215                    1360
...       ...           ...                       ...                     ...
339008      4            25                       990                    1171
339009      4            25                       680                     855
339010      4            25                       927                    1121
339011      4            25                       730                     804
339012      4            25                       995                    1199

        flight_number  scheduled_elapsed_time  departure_delay  distance  \
0                 746                    55.0             78.0       277
1                2126                   215.0              1.0      1670
2                  45                    95.0              9.0       580
3                  87                    95.0              1.0       580
4                 230                    85.0            108.0       580
...               ...                     ...              ...       ...
339008            319                   181.0             -6.0       998
339009           1719                   175.0             -4.0       998
339010            320                   134.0              4.0       689
339011            197                   194.0             -5.0      1222
339012           1086                   144.0             17.0       853

        cancelled  diverted  ...  dest_TXK  dest_TYR  dest_TYS  dest_VLD  \
0               0         0  ...     False     False     False     False
1               0         0  ...     False     False     False     False
2               0         0  ...     False     False     False     False
3               0         0  ...     False     False     False     False
4               0         0  ...     False     False     False     False
...           ...       ...  ...       ...       ...       ...       ...
339008          0         0  ...     False     False     False     False
339009          0         0  ...     False     False     False     False
339010          0         0  ...     False     False     False     False
339011          0         0  ...     False     False     False     False
339012          0         0  ...     False     False     False     False

        dest_VPS  dest_WRG  dest_XNA  dest_YAK  dest_YKM  dest_YUM
0          False     False     False     False     False     False
1          False     False     False     False     False     False
2          False     False     False     False     False     False
3          False     False     False     False     False     False
4          False     False     False     False     False     False
...          ...       ...       ...       ...       ...       ...
339008     False     False     False     False     False     False
339009     False     False     False     False     False     False
339010     False     False     False     False     False     False
339011     False     False     False     False     False     False
339012     False     False     False     False     False     False

[339013 rows x 618 columns]
```

--Train/TestSplit---

```python
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

-------------------------------------REGRESSION LINEAIRE---

```
model = LinearRegression()

model.fit(X_train, y_train)
```

---PREDICTIONS--------------------------------

```
y_pred = model.predict(X_test)
```

--------------------------------------MAE---------------------------

```
mae = mean_absolute_error(y_test, y_pred)

print(mae)
```

-------------------------------RMSE---

```
rmse = np.sqrt(mean_squared_error(y_test, y_pred))

print(rmse)
```

------------------------------Random Forest------------------------------

```
rf = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

rf.fit(X_train, y_train)
```